REPORT

OF

PASHUMITRA

A Ward-Level Stray Animal Grievance & Care Coordination Platform

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

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Batch: 2024-28

CANDIDATE’S DECLARATION

I, BHOOMI, do hereby declare that the internship report titled “PashuMitra — A Ward-Level Stray Animal Grievance & Care Coordination Platform” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: BHOOMI

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

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TABLE OF CONTENTS

3. PROJECT DESCRIPTION

3.1 Introduction

This report summarises a 4-week virtual internship undertaken in the role of Junior Web Developer, structured around a single continuous project: PashuMitra, a proposed e-governance digital service for municipal corporations. The internship was organised as a phased build-up — each week handed off directly into the next — moving from requirements analysis, through interface design, into a formal testing strategy, and finishing with a pre-launch audit against measurable performance, accessibility, and security benchmarks.

Rather than four disconnected assignments, the project was treated the way an actual product team would run a pre-development phase: define what to build and for whom, design how it should look and behave, define how it will be verified before launch, and finally audit a working build against that verification plan. A functional front-end prototype was also built independently to demonstrate the design in a real, clickable form.

3.2 Organization Profile

The supplied internship content does not identify a specific industry/organisation by name or provide a formal organisation profile. It describes the internship as a 4-week virtual internship in the role of Junior Web Developer and the project as a proposed e-governance digital service intended for deployment by an Indian municipal corporation.

3.3 Problem Statement

PashuMitra was designed around the need for a ward-level digital service that can coordinate stray-animal grievances and care activities rather than functioning as a generic citizen-complaint portal. The project is centred on the workflow described in the supplied report — capture, sterilisation, vaccination, and release or shelter placement — and on coordination among citizens, ward officers, AWBI-recognised NGOs, and veterinary teams.

The project therefore treats requirements, interface design, testing, accessibility, performance, and security as connected parts of one municipal service. The supplied report specifically notes that the work was grounded in existing initiatives and legal workflow rather than invented as a generic portal.

3.4 Project Objectives

- Apply web development and product-thinking skills to a realistic, socially relevant e-governance problem.
- Practice the full pre-build lifecycle of a digital product: requirements gathering, UX/UI design, QA strategy, and audit-based evaluation.
- Build and document professional-grade deliverables (project plans, design specifications, test strategies, audit reports) suitable for a real municipal stakeholder.
- Develop a working, testable prototype demonstrating the designed interface in functioning code.

3.5 Scope of the Project

PashuMitra ("friend of animals") is scoped as a ward-level stray animal grievance and care coordination platform for a municipal context. The supplied project description covers citizen grievance submission, case tracking, feeder registration, ward-level coordination, and a dashboard, while identifying citizens, ward officers, AWBI-recognised NGOs, and veterinary teams as key stakeholders.

The working prototype demonstrates the front-end scope described in the source content. It includes client-side form validation, a live case-tracking flow with a working status stepper, a feeder-registration flow that issues a working digital ID, and a dashboard that updates from live session data. The prototype is responsive across mobile and desktop widths.

The supplied report does not describe a production backend, database, or deployment infrastructure; the prototype was intentionally implemented as a front-end build using in-memory JavaScript state.

3.6 Technologies and Tools Used

Category	Skills / Tools Applied
Requirements & Planning	Stakeholder analysis, persona development, functional/non-functional requirement specification, risk analysis
UI/UX Design	Responsive/mobile-first design, fluid grid systems, WCAG 2.1 AA accessibility principles, wireframing

	and high-fidelity mockups
Front-end Development	HTML5, CSS3, vanilla JavaScript (DOM manipulation, form validation, client-side state management)
QA & Testing	Test pyramid design, test case writing, CI/CD pipeline concepts, risk-based test prioritisation
Performance & Security	Core Web Vitals, Lighthouse auditing, OWASP Top 10 vulnerability classes, accessibility auditing (axe-core, manual screen-reader testing)
Tooling & Version Control	Git and GitHub for version control and project hosting
Documentation	Structured technical report writing, data visualisation for audit/test findings, presentation design

3.7 System Architecture

The supplied content does not provide a formal system-architecture diagram. It describes the implemented prototype as a single-page application in vanilla HTML, CSS, and JavaScript, with in-memory JavaScript state used to simulate the flow from case submission to tracking and dashboard updates. The source therefore supports describing the prototype architecture at a high level as a client-side front-end with simulated state rather than a production multi-tier architecture.

The Week 1 grievance-lifecycle flowchart and Week 2 site map/user flow provide the available architectural/process-level representations in the supplied material.

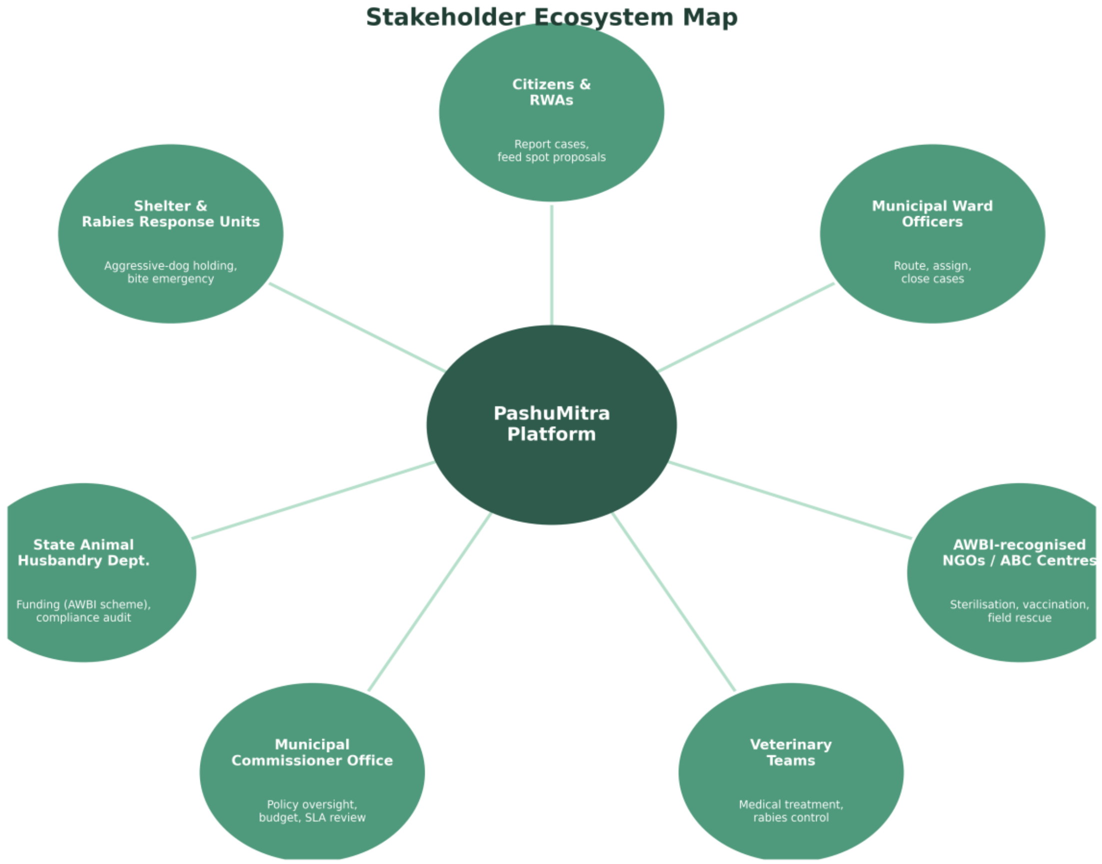

Figure 3.1 — Stakeholder ecosystem map produced in Week 1

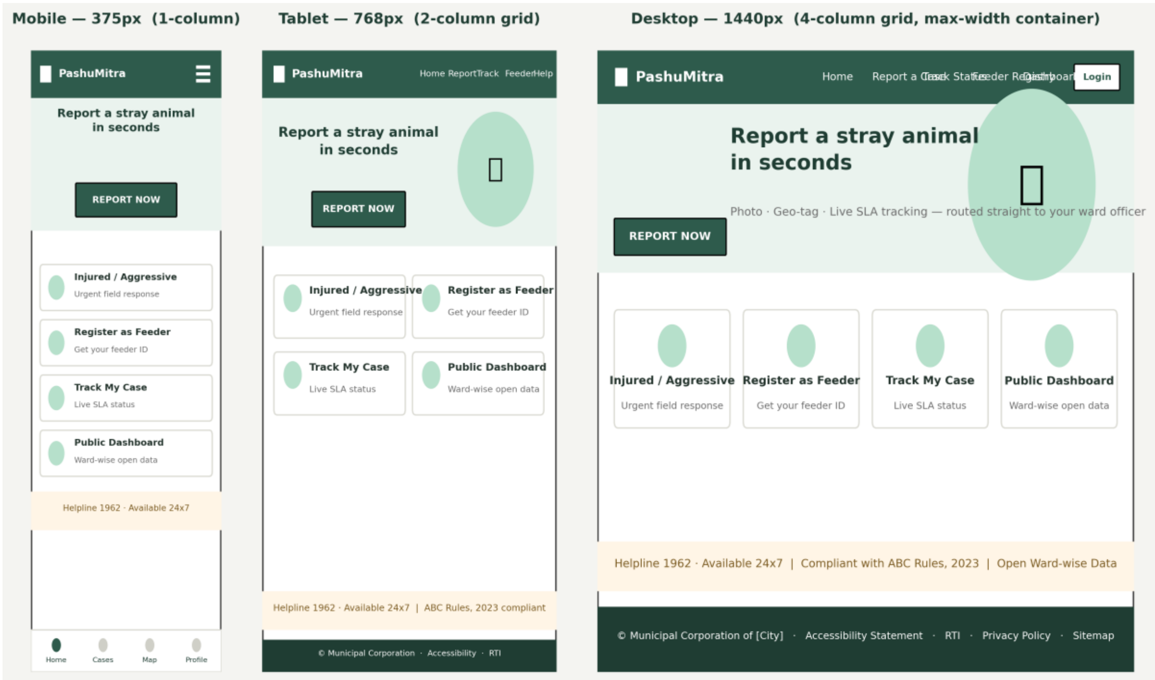

Figure 3.2 — Homepage design reflowing across mobile, tablet, and desktop breakpoints

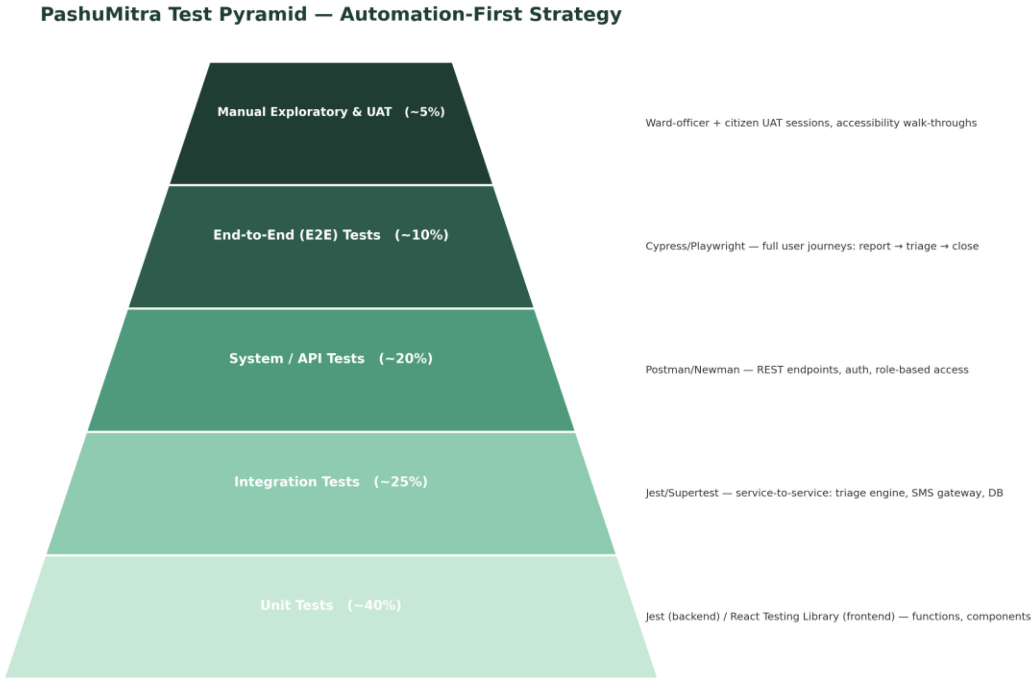

Figure 3.3 — Test pyramid defining the automation-first testing strategy

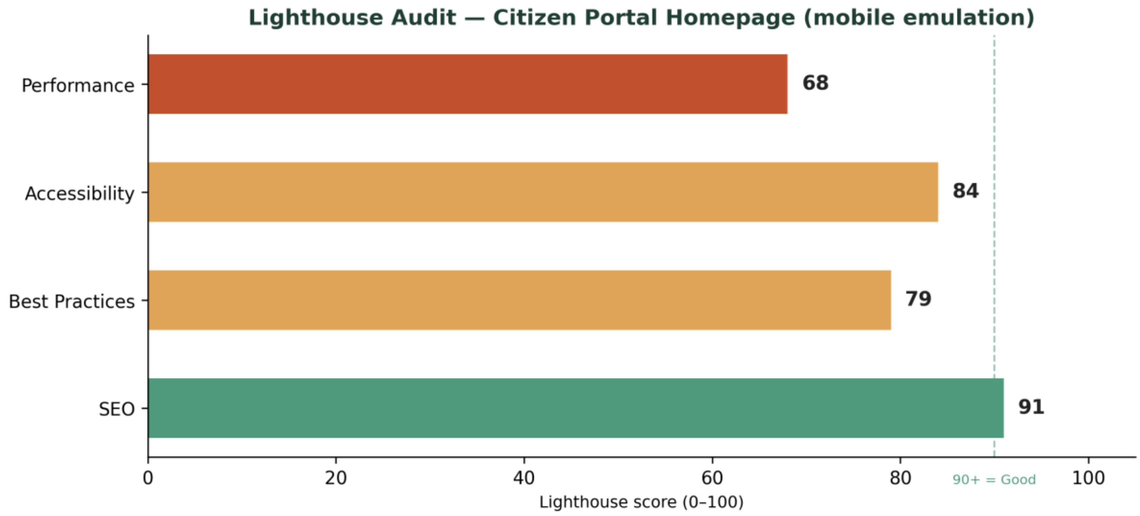

Figure 3.4 — Lighthouse audit scores for the citizen portal homepage

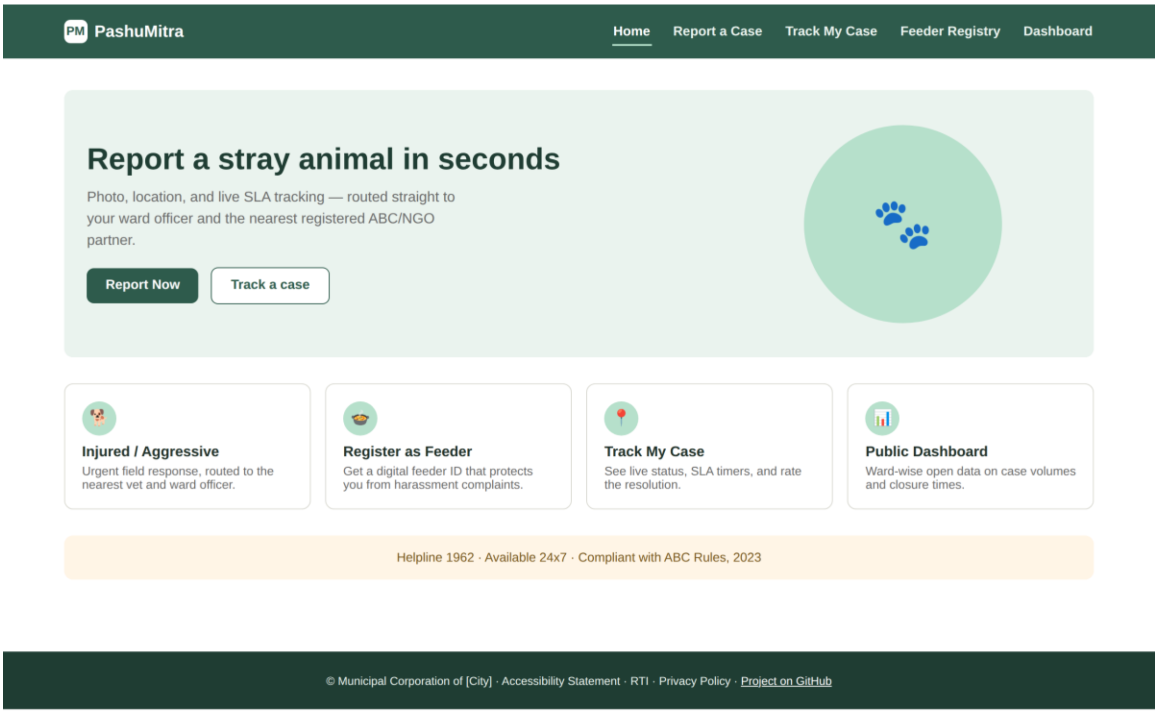

Figure 3.5 — Working prototype: the citizen portal homepage, rendered in-browser

3.8 Methodology

The internship followed a four-week continuous workflow in which each stage handed off directly into the next. The methodology moved from requirements and stakeholder analysis to interface design, then to QA/test planning, and finally to a performance, accessibility, and security audit.

Week 1 — Planning & Requirements Analysis

Began with research and benchmarking against real existing initiatives (Panvel's ABC Helpline, MCD's feeding-point registry, DigiLocker/UMANG identity patterns) to ground the plan in what municipal bodies are already attempting. Defined four stakeholder personas, 12 functional requirements, 9 non-functional requirements, and a full risk analysis, documented with a stakeholder ecosystem diagram, a grievance-lifecycle flowchart, and low-fidelity wireframes.

Week 2 — Responsive Web Prototype Design

Translated the Week 1 requirements into an actual interface: a full site map and user flow, and high-fidelity mockups of the homepage and report-submission form across three real breakpoints (375px mobile, 768px tablet, 1440px desktop). Design decisions — photo-upload placement, auto-detected ward field, privacy microcopy —

were each justified against a specific Week 1 requirement or persona pain point rather than chosen for visual preference alone.

Week 3 — QA & Testing Strategy

Defined a full test strategy organised as a test pyramid — unit, integration, system/API, end-to-end, performance, security, and accessibility testing — mapped to specific tooling (Jest, Cypress, k6, OWASP ZAP, axe-core) and a continuous testing pipeline where a failing stage blocks a pull request automatically. Seven detailed test cases were written with explicit pass/fail criteria, alongside a risk heatmap identifying the seven highest-priority risks with mitigation and contingency plans.

Week 4 — Performance, Accessibility & Security Audit

Audited a staging build of the citizen portal against Core Web Vitals, a full Lighthouse category breakdown, WCAG 2.1 AA accessibility criteria (organised by the POUR framework), and an OWASP-based security review. Findings included a critical broken-access-control vulnerability, stored XSS risk, and unmasked PII exposure — each paired with a concrete fix and prioritised P0/P1/P2 against a pilot-launch timeline.

Beyond the Weekly Deliverables — A Working Prototype

In addition to the four documentation deliverables, a functional front-end build was developed independently to demonstrate the design in working code rather than static mockups alone. Built as a single-page application in vanilla HTML, CSS, and JavaScript, it includes real client-side form validation, a live case-tracking flow with a working status stepper, a feeder-registration flow that issues a working digital ID, and a dashboard that updates from live session data — all fully responsive and tested at both mobile and desktop widths.

3.9 Expected Outcomes

The supplied content does not contain a separately labelled 'Expected Outcomes' section. The following outcomes are directly represented by the project's objectives and key learning outcomes: practical experience across the pre-development lifecycle; requirements-driven UX/UI decisions; risk-weighted QA and testing; familiarity with accessibility and security auditing; experience translating static designs into working front-end code; and improved technical documentation and presentation skills.

- Experience the full pre-development lifecycle of a digital product.
- Justify design decisions against specific user needs and requirements.
- Apply a risk-weighted approach to QA and testing.
- Read and interpret Lighthouse results, WCAG POUR principles, and common vulnerability classes such as broken access control and XSS.
- Translate static design into responsive front-end code using form validation, state handling, and layout techniques.
- Produce structured reports and presentations suitable for non-technical stakeholders.

Conclusion

Over four weeks, this internship moved from a blank requirements document to a fully designed, tested-on-paper, and audited digital service concept — backed by a working front-end prototype built independently to prove the design out in real code. The consistent thread across every deliverable was traceability: nothing was designed, tested, or audited in isolation from what the previous week had established. That discipline — carrying decisions forward instead of treating each week as a fresh assignment — is the main professional habit this internship reinforced, alongside the concrete technical skills in responsive design, QA strategy, and security/accessibility auditing.

5. BIBLIOGRAPHY / REFERENCES

The supplied completion report does not include a formal bibliography with publication details. It does identify the following project repository and the initiatives used during project research. No additional bibliographic metadata was supplied in the source document.

- PashuMitra project repository: https://github.com/bubiiiiiiiii/2024-2028_bhoomi_2410031616_5th_3cse25/tree/main
- Panvel ABC Helpline — identified in the supplied report as a benchmarking initiative.
- MCD feeding-point registry — identified in the supplied report as a benchmarking initiative.
- DigiLocker/UMANG identity patterns — identified in the supplied report as reference patterns.

- Animal Birth Control (ABC) Rules, 2023 — identified in the supplied report as the legal workflow context for the project.